From Econophysics to Quant

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Outline

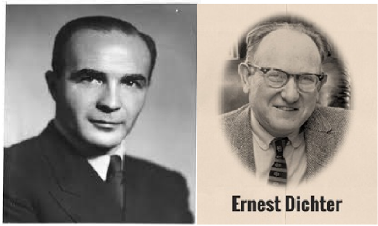
Motivation and Introduction

Challenger Disaster vs Richard Feynman

- Why Richard Feynman(Physicist and Noble lecturer)?
- A physicist is a natural philosopher of science who solves problems using mathematical models.

Ernest Dichter and Alfred Politz

- Ernest Dichter (14 August 1907 in Vienna – 21 November 1991 in Peekskill, New York) was an American psychologist and marketing expert known as the "father of motivational research. He has a **PhD in psychology.**
- Politz won the American Marketing Association's top leadership award in 1947 for developing more scientific standards and better techniques in marketing research. He has a **PhD in physics.**



Ernest Dichter and Alfred Politz

The New York Times

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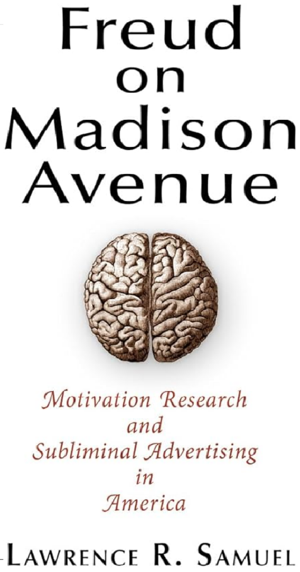
Alfred Politz, an early leader in the techniques of polling and opinion analysis and founder of a market research company, died of cancer Monday at his home in Odessa, Fla. He was 80 years old.

Mr. Politz established Alfred Politz Research Inc. in Manhattan in



What is the Quant?

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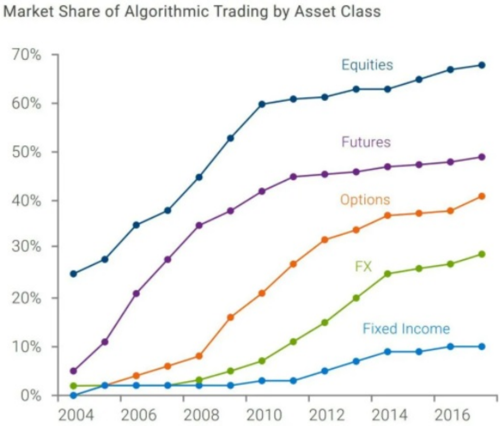
- Quqnt is research-based.
- Predicting the stock market.
- **Risk management** quantitative analysis or **"quants"**.
- The most successful hedge funds and financial management companies have employed quantitative methods, incorporating mathematics and physics.
- It may not come as a surprise that, until the year 2000, over one-third of the employees on Wall Street, including analysts, held Ph.D. degrees in physics.[1]

[1] The Dec 1999 issue of The Industrial Physicist magazine

What is the Quant?

- Algo Trading
- Mathematical Modeling and Risk Management
- Simulation
- Data analysis, designing approaches, and Back-testing the strategy
- Improving the trading architecture



Giants of Econophysics and Quant

Giants

- **Jim Simons**, *Mathematician(PhD in mathematics)*
  -Founder of Renaissance Technologies (one of the most successful (quantitative) hedges found in the world)
  -Faculty at Stony Brook University.
- **Didier Sornette**, *Physicist(PhD in physics)*
  -Founder of Risk-X
  -Director of the Financial Crisis Observatory
  -Co-founder of the ETH Risk Center
  -Professor of the ETH Zurich, UCLA, Swiss Finance Institute.
- **J. Doyne Farmer**, *Physicist(PhD in physics)*
  -Co-founder of Prediction Company (one of the first companies to do fully automated quantitative trading).
  -Professor of the Oxford University and the Santa Fe Institute.

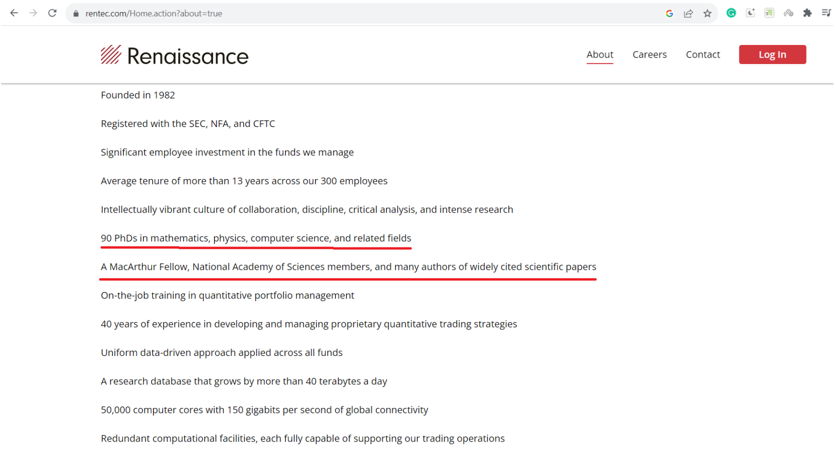
Giants

- **Jean-Philippe Bouchaud**, *Physicist(PhD in physics)*
  -Co-founder and chairman of Capital Fund Management (a global asset management company based in Paris with staff in New York City, London, and Sydney.)
  -Co-director of the CFM-Imperial Institute of Quantitative Finance at Imperial College London.
  -Adjunct professor at École Normale Supérieure.
- **Nigel Goldenfeld**, *Physicist(PhD in physics)*
  -Co-founder NumeriX(a company that develops high-performance software for pricing and risk managing derivative securities.)
  -Professor of the University of Illinois at Urbana-Champaign.
- **Rosario Nunzio Mantegna**, *Physicist(PhD in physics)*
  -Leader of New Tools in the Credit Network Modeling with Agents' Heterogeneity) at The Institute for New Economic Thinking (INET). -Professor of the Palermo University, the University College London, UCL, the Complexity Science Hub.

Big Institutes, Companies, Centers and Hedge Funds

Renaissance Technologies

- AUM US$130 billion
- Systematic trading using quantitative models derived from mathematical and statistical analysis.
- "Quants" with non-financial background.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Renaissance

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Founded in 1982

Registered with the SEC, NFA, and CFTC

Significant employee investment in the funds we manage

Average tenure of more than 13 years across our 300 employees

Intellectually vibrant culture of collaboration, discipline, critical analysis, and intense research

90 PhDs in mathematics, physics, computer science, and related fields

A MacArthur Fellow, National Academy of Sciences members, and many authors of widely cited scientific papers

On-the-job training in quantitative portfolio management

40 years of experience in developing and managing proprietary quantitative trading strategies

Uniform data-driven approach applied across all funds

A research database that grows by more than 40 terabytes a day

50,000 computer cores with 150 gigabits per second of global connectivity

Redundant computational facilities, each fully capable of supporting our trading operations

Prediction Company

- One of the first companies to do fully automated quantitative trading.
- The company used forecasting techniques to build black-box trading systems for financial markets, mainly employing statistical learning theory.
- In twenty-six years of operation, Prediction Company had only one losing year, 2007.

Numerix

- The company develops analytics and software for risk management, trading, valuations and pricing for both sell-side and buyside market participants.
- Numerix CrossAsset remains as one of the industry's most comprehensive collection of models offering full portfolio coverage and broad instrument support for fixed income/rates, equity, FX, credit, commodities, inflation, longevity, volatility and hybrids.
- Numerix peak revenue was $75.0M in 2023

Capital Fund Management

- CFM takes a scientific and academic approach to finance, using quantitative and systematic techniques to develop alternative investment strategies and products for institutional investors and financial advisers.
- CFM maintains strong links with academia.
- AUM \$10 billion

Institutes and Centers

- The Institute for New Economic Thinking at **University of Oxford**(UK)
- Santa Fe Institute(USA)
- Complexity Science Hub (Vienna)
- ECONOPHYSIX Capital Fund Management Chair of Econophysics & Complex Systems at **Ecole Polytechnique** (France)

The Quant is not the end!

The Quant is not the end!

- Using quantitative methods we are able to do many kinds of practical analysis.
- By developing mathematical models, simulations, and statistical analysis we can predict risks in **industry**, **banking supply chain**.
- We can use mathematical and statistical emergent approaches of Econophysics to analyze **real state price dynamics**!
- Using Agent-based models and complex system tools we can simulate future economic situations to have a clue of **the future**. (**do not compare them with AI!**)
- Predicts investment opportunity by mathematical analysis of the dynamics of that. For example, these people are strongly using methods to find the best investment opportunity in **renewable energy and technologies**. (Uni of Amsterdam, Oxford, NTNU)

What about us?

Thanks